Direct Competition between Working Memory Precision and Hippocampal Pattern Separation

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INTRODUCTION

Overall Aim

- Testing pattern separation hypo. for working memory precision
  - WM precision is supported by hippocampal pattern separation.



Working Memory Precision

- Correspondence between external stimuli & internal representations

Hippocampal Pattern Separation

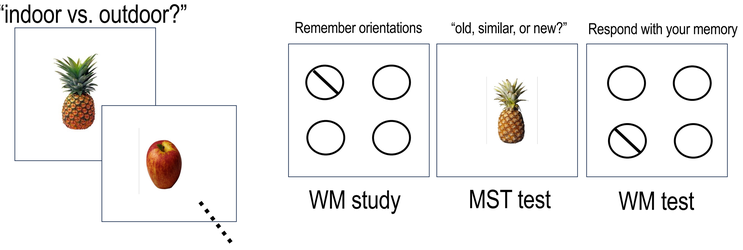
- Neurocomputational process of separating similar inputs into non-overlapping ones in the hippocampus[1]

Supporting Evidence

- WM representations of simple features decoded from hippocampal activity[2]
- Damage or stimulation on MTL impaired WM precision[3]

The present study

- Testing the direct competition between pattern separation and working memory precision



MST trial types

- Target – Same as the studied item
- Lure – Similar, but not the same (high pattern separation load)
- Foil – Not seen at study

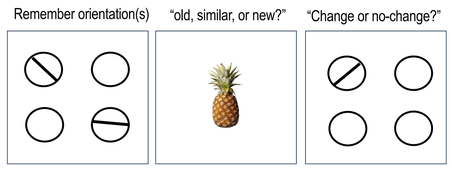
The Mixture model (slot model) [4, 5]

- WM retrieval are a mixture of likelihood of successful memory (capacity) and noisy representation (precision)

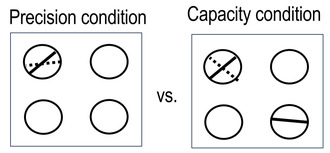
METHODS AND RESULTS

✓ Exp1: How does pattern separation demand affect WM performance (n = 47)

Change Detection task



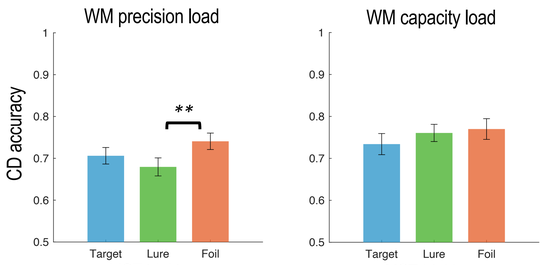
Manipulations

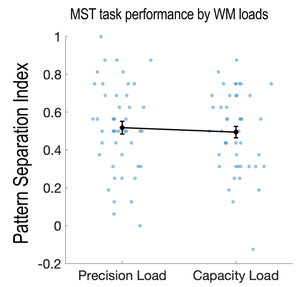


Set size 1 (small change) vs Set size 2 (large change)[6]

Prediction: CD performance will be reduced during MST lure trials, under WM precision condition

Results

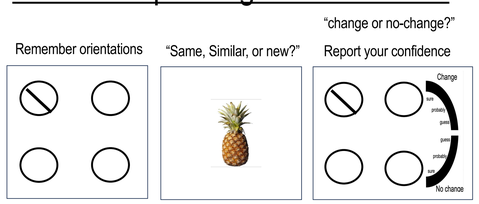




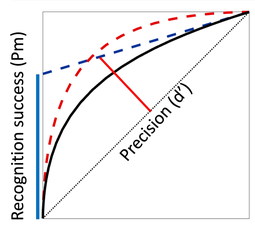
- Marginal two-way interaction (p = .056)
- Pairwise comparison result, $ACC_{lure} < ACC_{Foil}$
- No difference in PSI between WM loads
- No trade-off between WM and MST task performance

✓ Exp2 : Is direct measure of WM precision reduced during MST lures? (n = 44)

Receiver Operating Characteristics

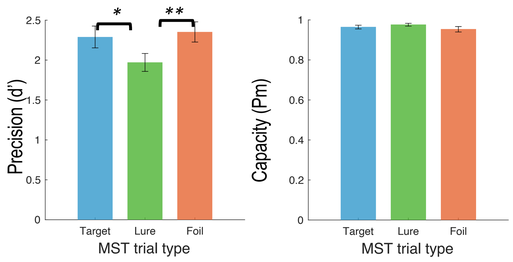


ROC Mixture Model[5]

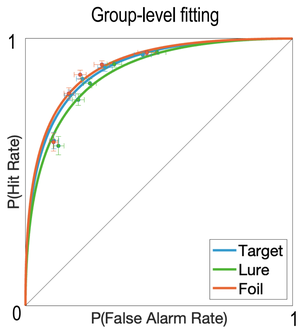


Prediction: WM precision will be reduced during MST lures, whereas capacity will be intact

Results



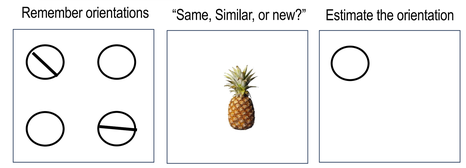
- Main effect of MST type on Precision (d')
- Pairwise comparison result, $d'_{lure} < d'_{Foil}$ & $d'_{lure} < d'_{Target}$
- No difference in Capacity (Pm) across MST trial type



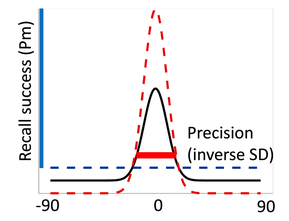
- Mean R^2 across subjects and conditions was 0.96

✓ Exp3 : WM precision is also reduced during MST lures using recall paradigm? (n = 44)

Recall task

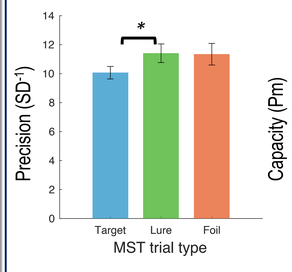


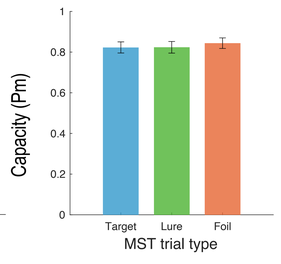
Recall Mixture Model[4]

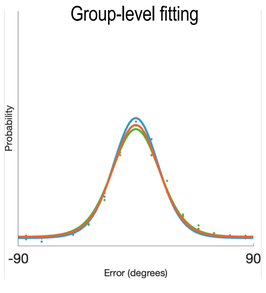


Prediction: WM precision will be reduced during MST lures

Results







- Main effect of MST type on Precision (SD^{-1})
- Pairwise comparison result, $SD_{lure} > SD_{Target}$
- No difference in Capacity (Pm) across MST trial type
- Mean R^2 across subjects and conditions was 0.81

✓ Discussion and Conclusions

- Reduced WM precision under high pattern separation load, but intact WM capacity, across three experiments
- Converging evidence from various methods supporting *hippocampal pattern separation hypothesis* of WM precision.

✓ Future studies

- Does perceptual discrimination process compete with WM precision?

✓ References

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2. Xie, W., Cappiello, M., Yassa, M. A., Ester, E., Zaghloul, K. A., & Zhang, W. (2023). The entorhinal-DG/CA3 pathway in the medial temporal lobe retains visual working memory of a simple surface feature. Elife, 12, e83365.
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4. Zhang, W., & Luck, S. J. (2008). Discrete fixed-resolution representations in visual working memory. Nature, 453(7192), 233–235.
5. Xie, W., & Zhang, W. (2017). Dissociations of the number and precision of visual short-term memory representations in change detection. Memory and Cognition, 45(8), 1423–1437.
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